Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/16/2025
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	FOAM 3; Augmentation of FOAM - Foam Optics And Mechanics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	03/26/2021	**End Date:**	03/25/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 03/25/2024 per D. Chao/GRC (Ed. 2/12/24). Original end date was 03/25/2023.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0898		
Performance Goal No.:			
Performance Goal Text:			

The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology.

Task Description:	There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies Earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into Advanced Life Support (ALS) technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character, in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability characteristics. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support (ALS). Water recovery and recycling are absolutely central to ALS, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems. The value of the FOAM project in terms of Earth benefits lies with a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for cleanup. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam.
Task Progress:	Ed. Note: The Principal Investigator has delivered a final report to NASA and provided the final summary below. For descriptions of the progress of FOAM investigations over time, please see the following investigations, and their Cumulative Bibliographies: . Grant 80NSSC21K0898: Foam 3; Augmentation of FOAM (Foam Optics and Mechanics) . Grant 80NSSC19K0599: Foam Optics and Mechanics . Grant NNX14AM99G: FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams) . Grant NNX07AP20G ; NNX07AK60G ; NNC04GB61G: The Melting of Aqueous Foams --- Aqueous foams are familiar from everyday life – from agitating soapy water, foaming up milk for cappuccino, etc. Microscopically, they consist of gas bubbles packed tightly into a relatively small amount of soapy liquid. Foams are important in many industries, both because of their unique mechanical properties and high interfacial area, and because they tend crop up and slow down processing of multi-component liquids. An important feature is that all foams evolve with time, by three distinct ways: (1) gravitational drainage of the liquid, (2) direct coalescence of neighboring bubbles by rupture/popping, and (3) the diffusion of gas out from smaller high-pressure bubbles into large low-pressure bubbles. The latter is called “coarsening” because it makes the average bubble size grow and causes the foam to have a coarser texture. While it’s possible to turn off (1) in microgravity, and (2) by good choice of surfactants or soaps, the coarsening mechanism is unavoidable. It can be slowed by using the least soluble gas possible, but it cannot be stopped. Unfortunately, the rate of coarsening is well understood only for the unrealistic / imaginary case of zero-wetness two-dimensional foams. One of the major motivating questions for the FOAM project is thus to elucidate the rate of coarsening for real three-dimensional foams and how it varies with foam wetness. This is difficult to answer because wetter foams drain faster, and because foams are opaque / white and we cannot simply peer into their interior because light reflects strongly from the multitude of gas-liquid interfaces. To make new progress on this old and important question, the FOAM project measured the coarsening process in microgravity for a series of foams with different degrees of wetness. Thus, it was possible to avoid the confounding effects of drainage that on Earth would obscure the nature of diffusive coarsening. To observe the coarsening process, the FOAM project both imaged surface bubbles and, very importantly, took advantage of the opaque nature of foams by measuring the fraction of incident light that diffuses through the samples, and relating this to bubble sizes. Thus, it was possible to characterize the bubble-scale structure in the interior and study how it evolved with time. This was done for foams with different degrees of wetness and for foams with different kinds of surfactants. The data collected by the FOAM project provided a new and deeper understanding of how the rate of coarsening depends upon both liquid content and chemical composition. This was aided by construction of theoretical models of the coarsening process and testing / comparing them against the data. The new understanding is important not just for pure / fundamental reasons, but also because the coarsening process can then be better controlled in their many industrial applications. In addition, the creation of high-performance metallic alloys and liquid emulsions both depend upon an analogous coarsening process, and our new insights can be transferred.

Bibliography Type:	Description: (Last Updated: 06/12/2025)
Articles in Other Journals or Periodicals	Durian DJ. "Effective exponents for the diffusive coarsening of wet foams and analogous materials." arXiv preprint server. Posted April 2, 2023. https://doi.org/10.48550/arXiv.2304.00415 , Apr-2023
Articles in Peer-reviewed Journals	Dillavou S, Hanlan JM, Chieco AT, Xiao H, Fulco S, Turner KT, Durian DJ. "Bellybutton: accessible and customizable deep-learning image segmentation." Scientific Reports. 2024 Jun 20;14(1):14281. https://doi.org/10.1038/s41598-024-63906-y , Jun-2024
Articles in Peer-reviewed Journals	Pasquet M, Galvani N, Pitois O, Cohen-Addad S, Höhler R, Chieco AT, Dillavou S, Hanlan JM, Durian DJ, Rio E, Salonen A. "Aqueous foams in microgravity, measuring bubble sizes." Comptes Rendus. Mécanique. 2023;351(S2):139-61. http://dx.doi.org/10.5802/crmeca.153 , May-2023